

WEEK 4 — Advanced Network Forensic Challenges & Mini Investigation

HERNANDO, PANTINE B.

BSIT-3

Overview

In Week 4, I performed advanced network forensic challenges using PCAP files and analyzed them using **Wireshark**. The goal was to investigate network traffic, reconstruct communication streams, and extract hidden flags from captured packets. The activity simulated a mini forensic investigation where network evidence must be examined to uncover hidden information.

Investigation Experience

During the analysis, I inspected different protocols such as TCP, UDP, ARP, and other background traffic. Many packets initially appeared as normal network activity, which made it challenging to determine which packets contained relevant data.

For **Challenge 6**, the flag was hidden within TCP packets where each character was separated by spaces. By using Wireshark's **Follow TCP Stream** feature, I reconstructed the communication stream and extracted the hidden message. The recovered flag was:

picoCTF{p4ck37_5h4rk_ceccaa7f}

For **Challenge 7**, the PCAP file contained multiple protocols and network noise. It was difficult to identify the correct packet at first. After filtering the traffic and analyzing UDP communication, I used the **Follow UDP Stream** feature and successfully located the hidden data in the stream. The recovered flag was:

picoCTF{StaT31355_636f6e6e}

Challenges Encountered

One of the main challenges during the investigation was distinguishing meaningful packets from normal network traffic. Many packets contained broadcast messages and unrelated activity, which required careful filtering and analysis.

Another difficulty was understanding how the data was formatted inside the packets. In some cases, the flag was fragmented or spaced out, which required reconstructing the stream and cleaning the output before identifying the final flag.

Conclusion

This activity strengthened my skills in network forensic analysis using Wireshark. I learned how to analyze packet captures, filter protocols, reconstruct TCP and UDP streams, and extract hidden information from network traffic. The challenges also improved my ability to investigate suspicious network behavior and perform a basic DFIR-style investigation.